



Write MATLAB code to evaluate the softmax function at given points

The softmax function is presented below.

$$y_i(\mathbf{x}) = \frac{e^{x_i}}{\sum_{j=1}^K e^{x_j}} \text{ for } i = 1, \dots, K \text{ and } \mathbf{x} = (x_1, x_2, \dots, x_K) \in \mathbb{R}^K$$

The softmax function takes a vector x with K real numbers as input. The output is a vector y containing K real numbers, where the sum of all elements in y equals 1. This function is useful for mapping real numbers to probabilities and is widely used in machine learning. You are provided with a 4×3 matrix, where each row represents a vector of $K = 3$ real numbers. For each row vector, compute the corresponding softmax output vector. The output should also be a 4×3 matrix, where each row is the softmax result of the corresponding input row vector. It is recommended to use vectorized operations to process entire rows at once.

Although you can use the way in the warm-up exercise to get the full mark, can you use a single line of code to achieve this task?

Script

Save

Reset

MATLAB Documentation (https://www.mathworks.com/help/)

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```
1 % Apply softmax function to every row.
2 x = [15 , -1 , 8 ;
3      -10 , 20 , 0 ;
4      15, 0.5, 8;
5      0.2, 10 , 0.5 ];
6 % y is the output.
7 y = (exp(x') ./ sum(exp(x')))' ;
8
```

Run Script

Tests: All Tests Passed

Submit

- ✔ Do not use the built in SoftMax function.
- ✔ Are exponential values being calculated appropriately?
- ✔ Is the sum being calculated appropriately?
- ✔ Is the output correct?

Output

Code ran without output.